



# Complete Vegetable Storage & Intelligence Report

Reporting Period: 24h

OPERATOR: Dr. Aris Thorne (demo@vegsense.io)

DEVICE: ESP32-DEMO-001 [DEMO DATA]

TARGET BATCH: BATCH-2026-TOM-01 (12 kg)

TELEMETRY: 24 points (raw)

## Executive Environmental Summary

AVG TEMPERATURE

**27.6°C**

Range: 26.2 - 29°C

AVG HUMIDITY

**73.3%**

Range: 70 - 76%

AVG GAS/VOC INDEX

**419**

Relative MQ-135 index

AVG LIGHT LEVEL

**178 lx**

LOW LIGHT

ESTIMATED RISK

**17%**

Status: FRESH

TOTAL ALERTS

**9**

0 critical, 3 active

MONITORED MASS

**82 kg**

0 active batches

DATA QUALITY

**85%**

FAIR

## Atmospheric Sensor Monitoring Analysis

Historical statistics recorded across storage chamber sensor arrays.

| SENSOR METRIC         | CURRENT | AVERAGE | MINIMUM | MAXIMUM | CONDITION STATUS |
|-----------------------|---------|---------|---------|---------|------------------|
| Temperature (°C)      | 26.6°C  | 27.6°C  | 26.2°C  | 29°C    | NORMAL           |
| Relative Humidity (%) | 73%     | 73.3%   | 70%     | 76%     | OPTIMAL          |
| Gas/VOC Indicator     | 412     | 419     | 411     | 431     | NORMAL           |
| Light Level (lux)     | 0 lx    | 178 lx  | 0 lx    | 450 lx  | LOW LIGHT        |

## Estimated Spoilage Risk & Factor Breakdown

Multivariate risk scoring derived from Part 6 preservation heuristics.

RISK INDEX SUMMARY:

Current: 16% | Average: 17% | Classification: FRESH | Historical Trend: STABLE

FACTOR CONTRIBUTIONS:

Thermal Impact: 15% | Moisture Impact: 14% | VOC Impact: 18% | Photometric: 10%

RISK SPECTRUM TIME:

Fresh (0–30%): 100% | Warning (31–60%): 0% | Risk (61–80%): 0% | Critical: 0%





Alert Notifications & Atmospheric Incidents

Total recorded: 9 (Critical: 0, High: 5, Warning: 2) | Avg Resolution: 3 min

| Timestamp | Incident Title                  | Category                | Severity | Status   | Device         |
|-----------|---------------------------------|-------------------------|----------|----------|----------------|
| 01:16 pm  | Gas/VOC Sensor Data Unavailable | SENSOR_DATA_UNAVAILABLE | WARNING  | RESOLVED | ESP32-DEMO-001 |
| 01:16 pm  | Light Sensor Data Unavailable   | SENSOR_DATA_UNAVAILABLE | WARNING  | RESOLVED | ESP32-DEMO-001 |
| 01:16 pm  | High Humidity                   | HUMIDITY                | HIGH     | RESOLVED | ESP32-DEMO-001 |
| 01:17 pm  | Gas/VOC Sensor Recovered        | SENSOR_RECOVERY         | INFO     | ACTIVE   | ESP32-DEMO-001 |
| 01:17 pm  | Light Sensor Recovered          | SENSOR_RECOVERY         | INFO     | ACTIVE   | ESP32-DEMO-001 |
| 01:17 pm  | High Humidity                   | HUMIDITY                | HIGH     | RESOLVED | ESP32-DEMO-001 |
| 01:18 pm  | High Humidity                   | HUMIDITY                | HIGH     | RESOLVED | ESP32-DEMO-001 |
| 01:29 pm  | High Humidity                   | HUMIDITY                | HIGH     | RESOLVED | ESP32-DEMO-001 |

Vegetable Storage Inventory Status

Total inventory: 4 batches (82 kg) across 4 produce types.

| Produce Variety | Batches     | Estimated Risk | Alerts | Storage Condition |
|-----------------|-------------|----------------|--------|-------------------|
| Tomato          | 1 batch(es) | 18%            | 0      | OPTIMAL           |
| Potato          | 1 batch(es) | 12%            | 0      | OPTIMAL           |
| Onion           | 1 batch(es) | 14%            | 0      | OPTIMAL           |
| Carrot          | 1 batch(es) | 19%            | 0      | OPTIMAL           |

Actionable Storage & Engineering Recommendations

Targeted adjustments derived from active telemetry and threshold events.

ACTIVE ALERTS

Acknowledge and inspect storage bays corresponding to active alert triggers.

3 unresolved incident notifications currently flagged.

TECHNICAL DISCLAIMER & SYSTEM SCOPE NOTE

VegSense provides environmental monitoring and estimated storage-condition/spoilage risk based on available sensor and storage data. The Gas/VOC sensor is an environmental indicator and does not by itself identify a specific spoilage gas. Risk classifications are configurable estimates and should not be interpreted as definitive confirmation of physical spoilage.

